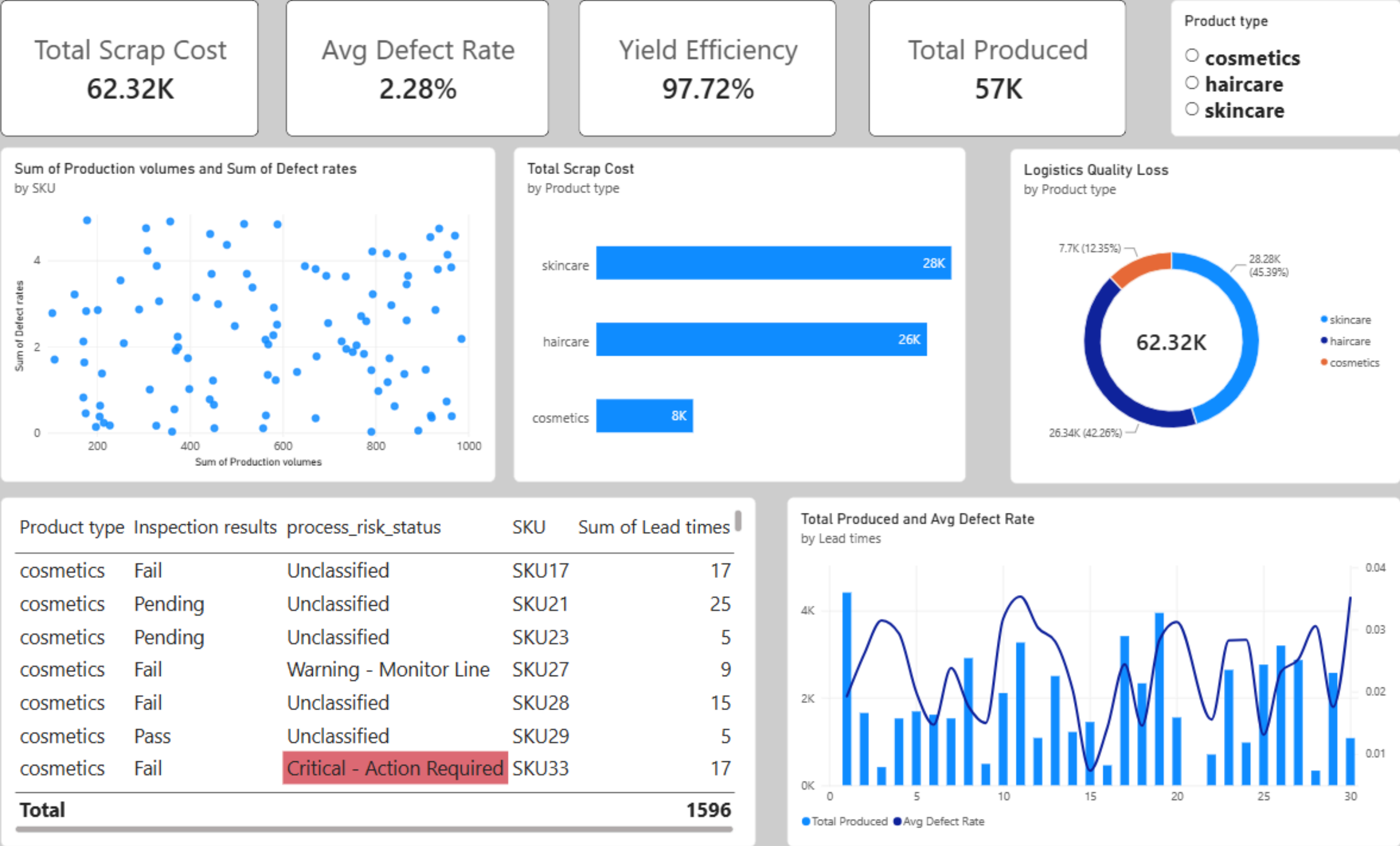


MANUFACTURING QUALITY & PROCESS RISK OPTIMIZATION ENGINE

Reducing Operational Waste and Scrapped Batches using Data Analytics



The main goal of this project was to isolate manufacturing quality bottlenecks, quantify the exact financial cost of defects, and provide live, actionable tracking for plant managers.

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The Cost of Visibility Gaps on the Factory Floor

- **Unquantified Scrap:** Quality and production teams manually track failures, but do not look at the holistic, cumulative financial waste.
- **Reactive Post-Mortems:** Defect trends are analyzed weeks after a production run has ended, resulting in continuous material loss.
- **Target Scope:** Pinpoint exactly which batches, SKUs, and product lines are bleeding operational capital

Pipeline Architecture & Feature Engineering

- **Data Filtration:** Cleaned and isolated critical production lines (57,000+ units across Cosmetics, Haircare, Skincare).
- **Metric Formulation:** Engineered the Cost of Poor Quality (COPQ) tracking calculation in Pandas:
$$\text{COPQ} = \text{Defective volume} \times \text{unit manufacturing cost}$$
- **Risk Categorization:** Programmed a predictive Process Risk Status logical matrix using NumPy conditional evaluations to dynamically segment batches into Optimal, Warning, or Critical triage alerts.

Uncovering the Core Manufacturing Leaks

- **Baseline Performance:** Factory floor operations currently maintain a 97.72% First-Time Yield (FTY) and a 2.28% Average Defect Rate.
- **The 80/20 Rule:** Financial waste is highly concentrated. Over 80% of total scrap costs (\$62.32K) originate in the Skincare and Haircare divisions, while Cosmetics remains highly controlled.
- **Batch Volatility:** The scatter plot distribution indicates a sharp inflection point where large-volume runs exhibit higher variance in quality control metrics.

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Dashboard

Continuous Improvement Action Plan

- **Targeted Machine Interventions:** Prioritize preventive maintenance routines specifically on Skincare manufacturing tooling to flatten the high defect trends.
- **Batch Size Standardization:** Optimize scheduling limits to avoid pushing high-volume runs past the verified quality threshold.
- **Estimated Savings:** Implementing these data-driven process safeguards stands to recapture up to 15–20% of the currently lost \$62.32K in annual waste.

THANK YOU